L20 — Circular Linked List and Circular Doubly Linked List

Unit: 2 | **Chapter:** Linked Lists | **BT Level:** BT3 | **CO:** CO2

Learning Objectives

- Explain the structure of circular singly and doubly linked lists
- Implement insertion, deletion, and traversal for circular linked lists
- Identify use cases where circular lists are preferred
- Differentiate circular lists from their linear counterparts

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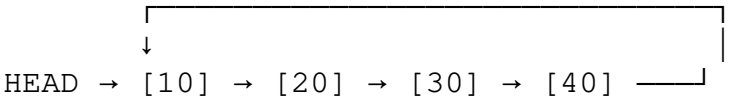
1. Circular Singly Linked List (CSLL)

In a **circular singly linked list**, the last node's `next` pointer points back to the **HEAD** (not NULL).



There is **no NULL** in the chain — the list wraps around.

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2. CSLL Operations

2.1 Traversal — must detect when we're back at HEAD

```
def traverse_csll(head):
    if head is None:
        return
    current = head
    while True:
        print(current.data, end=" → ")
        current = current.next
        if current == head:    # Full circle completed
            break
    print("(back to head)")
```

2.2 Insert at Beginning — O(n) (need to update last node)

```
def insert_beginning(self, data):
    new_node = Node(data)
    if self.head is None:
        new_node.next = new_node   # Points to itself
        self.head = new_node
```

```
        return

    # Find last node (the one pointing to head)
    last = self.head
    while last.next != self.head:
        last = last.next

    new_node.next = self.head
    last.next = new_node     # Last node now points to new head
    self.head = new_node
```

2.3 Insert at End — O(n)

```
def insert_end(self, data):
    new_node = Node(data)
    if self.head is None:
        new_node.next = new_node
        self.head = new_node
        return

    last = self.head
    while last.next != self.head:
        last = last.next

    last.next = new_node
    new_node.next = self.head   # New node circles back to head
```

2.4 CSLL with Tail Pointer — O(1) insert at end

Using a **tail pointer** (instead of head) avoids O(n) traversal:

```
class CSLL_WithTail:
    def __init__(self):
        self.tail = None    # Points to last node

    def insert_end(self, data):
        new_node = Node(data)
        if self.tail is None:
            new_node.next = new_node
            self.tail = new_node
        else:
            new_node.next = self.tail.next   # New last → old first (he
            self.tail.next = new_node
            self.tail = new_node

    @property
    def head(self):
        return self.tail.next if self.tail else None
```

With tail pointer: insert at end → O(1), insert at beginning → O(1).

2.5 Delete a Node — O(n)

```
def delete_node(self, key):
    if self.head is None:
        return

    last = self.head
    while last.next != self.head:
        last = last.next

    # Deleting head
    if self.head.data == key:
        if self.head == last:    # Only one node
            self.head = None
        else:
            last.next = self.head.next
            self.head = self.head.next
        return

    # Search and delete non-head node
    curr = self.head
    while curr.next != self.head:
        if curr.next.data == key:
            curr.next = curr.next.next
            return
        curr = curr.next
```

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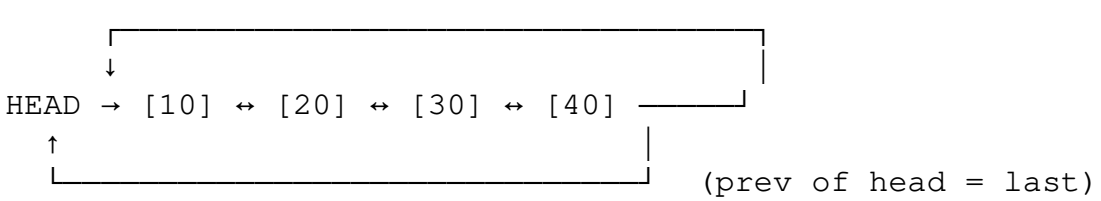
3. Circular Doubly Linked List (CDLL)

In a **CDLL**, each node has both `prev` and `next` pointers, and:

- last.next = first
- first.prev = last



3.1 Node Structure

```
class CDLLNode:
    def __init__(self, data):
        self.data = data
        self.prev = None
        self.next = None
```

3.2 CDLL with Sentinel (Header) Node

The cleanest implementation uses a **sentinel/dummy node**:

```
class CDLL:
    def __init__(self):
```

```
        self.sentinel = CDLLNode(None)   # Dummy node
        self.sentinel.next = self.sentinel
        self.sentinel.prev = self.sentinel

    def insert_after(self, node, data):
        """Insert new node after 'node' — O(1)"""
        new_node = CDLLNode(data)
        new_node.next = node.next
        new_node.prev = node
        node.next.prev = new_node
        node.next = new_node

    def insert_front(self, data):
        self.insert_after(self.sentinel, data)

    def insert_back(self, data):
        self.insert_after(self.sentinel.prev, data)

    def delete_node(self, node):
        """Remove a node — O(1) since we have prev pointer"""
        node.prev.next = node.next
        node.next.prev = node.prev

    def traverse_forward(self):
        curr = self.sentinel.next
        while curr != self.sentinel:
            print(curr.data, end=" ⇔ ")
            curr = curr.next
        print("(sentinel)")

    def traverse_backward(self):
        curr = self.sentinel.prev
        while curr != self.sentinel:
            print(curr.data, end=" ⇔ ")
            curr = curr.prev
        print("(sentinel)")

    def is_empty(self):
        return self.sentinel.next == self.sentinel
```

3.3 Demo

```
cdll = CDLL()
cdll.insert_back(10)
cdll.insert_back(20)
cdll.insert_back(30)
cdll.insert_front(5)
cdll.traverse_forward()    # 5 ⇔ 10 ⇔ 20 ⇔ 30 ⇔ (sentinel)
cdll.traverse_backward()   # 30 ⇔ 20 ⇔ 10 ⇔ 5 ⇔ (sentinel)
```

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4. Complexity Comparison

Operation	CSLL (with tail ptr)	CDLL (with sentinel)
Insert front	O(1)	O(1)
Insert back	O(1)	O(1)
Delete given node	O(n)	**O(1)**
Traverse forward	O(n)	O(n)
Traverse backward	Not possible	O(n)

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5. Real-World Applications of Circular Lists

Application	Why Circular?
Round-robin scheduler	Each process gets CPU turn in cycles
Audio/video buffer	Circular buffer (ring buffer) for streaming
Board games (Monopoly)	Players take turns in a cycle
Browser tabs	Ctrl + Tab cycles through tabs
Multiplayer games	Turn rotation among players
Music player	Looping playlist

Ring Buffer (Circular Queue) — O(1) enqueue and dequeue

```
class RingBuffer:
    def __init__(self, capacity):
        self.buffer = [None] * capacity
        self.capacity = capacity
        self.head = 0
        self.tail = 0
        self.size = 0

    def enqueue(self, item):
        if self.size == self.capacity:
            raise Exception("Buffer full")
        self.buffer[self.tail] = item
        self.tail = (self.tail + 1) % self.capacity
        self.size += 1

    def dequeue(self):
        if self.size == 0:
            raise Exception("Buffer empty")
        item = self.buffer[self.head]
        self.head = (self.head + 1) % self.capacity
        self.size -= 1
        return item
```

Summary

- **Circular Singly Linked List:** last.next → HEAD; useful for round-robin applications.
- With a **tail pointer**, both front and back insertions become O(1).

- **Circular Doubly Linked List:** combines CDLL bidirectionality with circular structure.
- A **sentinel node** simplifies implementation — no NULL checks needed.
- CDLL with sentinel: O(1) insert/delete anywhere given node reference.

References

- Cormen et al., *Introduction to Algorithms*, Ch. 10.2 (MIT Press, 2022)
- Skiena, *The Algorithm Design Manual*, Ch. 3.1 (Springer, 2020)